

Lifting the Landscape: How Within-Lifetime Learning Affects Evolution in an ALife Environment

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Abstract

The relationship between learning and evolution remains poorly understood in biological contexts, and is also not obvious as inheritance is non-Lamarckian. Evolutionary theories, such as the Baldwin Effect, suggest that within-lifetime learning can increase fitness as organisms adapt to environments, speeding up behavioural encoding. In this paper, we use an agent-based ALife simulation called Life Engine to investigate the effects of within-lifetime learning on evolution and behavioural encoding. Using an evolutionary algorithm for evolution and reinforcement learning for within-lifetime learning, we compare organisms in different environments of varying fitness landscape difficulty to show that within-lifetime learning is beneficial on smooth yet more difficult fitness landscapes, reliably and consistently climbing the fitness landscape. We also show that within-lifetime learning can reshape the fitness landscape by lifting it into regions of higher fitness. From a biologically inspired investigation, this suggests that evolution is sufficient on smooth non-difficult fitness landscapes, but that within-lifetime learning helps on more difficult and complex landscapes, which are found in nature.

Keywords

Artificial life, within-lifetime learning, evolutionary algorithms, fitness landscape, reinforcement learning

1 Introduction

Behaviour plays an important role in shaping evolution, with beneficial behaviours being preserved by natural selection [13, 18]. Although genes do not specify what behaviours organisms have, they do shape the expression of behaviours, which has led to theories that behaviours can become genetically assimilated [20, 27]. One important theory that suggests this is the Baldwin Effect, which posits that learned behaviours eventually become genetically encoded if beneficial to survival [2].

Learned behaviours may seem unrelated to behavioural encoding as they are non-heritable, but the Baldwin Effect suggests that as organisms learn throughout their lifetime, their fitness can increase. As fitness is a measure of an organism's ability to survive and reproduce, fitness-increasing behaviours are selected for [21]. This reasoning grounds learning as an important mechanism for evolution, yet the relationship between the two remains poorly understood because of the lack of empirical evidence. There have been computational and theoretical studies of the effects of within-lifetime learning on evolution and genetic encoding, which have considered extreme fitness landscapes and simple tasks, but biologically inspired empirical evidence is still necessary [24].

Studying this relationship could require up to millions of years on evolutionary time-scale given the slow rate of evolution, making Artificial Life (ALife) a suitable medium. The field of ALife can be defined as "a scientific discipline that seeks to study and understand the principles of life in order to create artificial systems that synthesize natural life and transnatural processes, oriented toward a specific purpose." [25], and is thus useful to study processes like evolution. Soft ALife, in particular, can be used to study processes that would take millions of years in the natural world, as life is synthesised in a computer simulation that can simulate this time in hours or days [1]. Agent-based soft ALife simulations more realistically model the natural world as each agent, or digital organism, has some level of autonomy leading to emergent behaviours and interaction within the environment [6].

Using Life Engine ¹, an open source agent-based ALife simulation, we investigate the relationship between learning and evolution in different environments. Using this simulator, we study the effects of within-lifetime learning as it relates to the Baldwin effect and the evolution of viable genomes, which allow a population to successfully reproduce. In our experiments, viable genomes are genomes above a particular fitness threshold defined in Section 4.2. We investigate the research questions:

- RQ1. Does within-lifetime reinforcement learning improve a population's ability to reach and sustain viability relative to evolutionary search alone?
- RQ2. Does within-lifetime reinforcement learning help with behavioural encoding of competent food-seeking behaviour across generations of digital organisms, as predicted by the Baldwin Effect?

These research questions emerged when investigating existing literature on within-lifetime learning. Hinton & Nowlan claimed that learning guided evolution using an extreme fitness landscape with only one peak and one perfect solution [9]. This fitness landscape is not representative of those found in nature, which motivates using ALife as a medium and our research questions. ??????

FOOD SEEKING BEHAVIOUR TO MEASURE FITNESS>?????

We use multi-layer-perceptrons (MLPs) [22] serving as genomes, reinforcement learning (RL) [23] and an evolutionary algorithm (EA) [5] to simulate within-lifetime learning and evolution in different environments with varying difficulty. Our independent variables are the adaptation mechanism (EA, RL + EA or only RL) and the environment (baseline or predator). Our dependent variables are the normalised mean fitness, runs that exceed the viability threshold and how the fitness landscape slices change in the presence of learning.

¹<https://thelifeengine.net/>

Using the three independent variables, we isolate the effects of within-lifetime and selection mechanisms, helping to pinpoint when within-lifetime is effective. The baseline environment has simple rules, allowing for competent food-seeking behaviour to be easily encoded, and the predator environment has predators that add difficulty to the fitness landscape. From these, we hypothesise that the presence of learning will be beneficial in both environments and that learning needs a selection mechanism to be useful. We also hypothesise that evolution encodes behaviours less quickly than learning.

From this design, we show that within-lifetime learning can be a beneficial and robust mechanism when the fitness landscape is difficult, reliably helping a population to reach viability when a selection mechanism is present. We also show that in the presence of a selection mechanism, within-lifetime learning can reshape the fitness landscape by lifting it into regions of higher fitness.

2 Related Work

This section provides an overview of relevant literature to our experiments and research questions. We begin with a review of fitness landscapes and how they are classified as classification is necessary for our methodology, and reviewing theoretical models that have considered the effects of within-lifetime learning. We also consider the specifics of the claims made by the Baldwin Effect and close with a review of how Genetic Algorithms have been used in combination with learning, which are necessary for our research questions and metrics.

2.1 Fitness Landscapes

A fitness landscape is an abstract space that represents genomes with different fitness values, creating a landscape with contours, peaks and valleys that show regions of different fitness [30]. Evolution climbs the fitness landscape, reaching regions with higher fitness values as natural selection allows for genomes to become fitter over time. Within this abstract representation, there are countless possibilities for different problems and environments, so researchers have developed ways to classify and describe landscapes.

The most fundamental classifications are smooth and rugged landscapes. Smooth landscapes have similar fitness scores with large basins of attraction where the selection signal is clear to allow a gradual improvement [26]. A rugged landscape is one where neighbouring genomes have large differences in fitness values and one with many local optima [15]. Previous studies that have investigated the interaction between learning and evolution have classified fitness landscapes as smooth or rugged [24].

As domains other than biology have also used the notion of a fitness landscape to classify and describe problems, attributes such as modality and difficulty have been introduced to more accurately describe beyond smooth and rugged [15].

2.2 The Baldwin Effect

The Baldwin Effect suggests that phenotypic plasticity accelerates genetic encoding [2]. The reasoning is that organisms that can respond to environmental changes reach higher fitness levels than organisms who do not, allowing them to be favoured by

natural selection. Despite these learned behaviours following non-Lamarckian inheritance by being non-heritable, genetic makeup that allows for this plasticity is preserved, gradually encoding these beneficial behaviours. According to Baldwin, within-lifetime learning shields organisms from environmental changes and accelerates genetic encoding because phenotypic plasticity allows for faster adaptation than mutation [24].

The Baldwin Effect is an important theory which suggests that behaviour and learning have an impact on evolution, but biological evidence is missing. There have been various computational and theoretical studies about the Baldwin Effect, but there have been contradictory conclusions where learning either accelerates or decelerates evolution [24].

2.3 Computational & Theoretical Models of Learning and Evolution

The foundational paper by Hinton & Nowlan investigate the Baldwin Effect in a simulation environment, concluding that learning has a smoothing effect on the fitness landscape by reducing the search space, which evolutionary search cannot do [9]. This work uses an extreme "needle-in-a-haystack" fitness landscape with no selection signal, which shows the benefits of learning, but is unrealistic in the natural world as there is only one correct solution.

Works by Mayley and Borenstein further investigate the effects of learning on evolution in computational and theoretical studies respectively, suggesting that learning has a cost and that the effectiveness of learning depends on the number of peaks and valleys on the fitness landscape i.e. ruggedness [4, 17]. Although these works offer more insight into the effects of learning on different fitness landscapes, they remain theoretical and do not consider interactions between organisms and the environment as found in the natural world. These works also only differentiate between landscapes in terms of smoothness and ruggedness, not considering attributes such as difficulty, which can exist on smooth landscapes [15].

2.4 Learning Algorithms & Genetic Algorithms

Gu et al. shows the effectiveness of combining Deep Reinforcement Learning (DRL) and a genetic algorithm (GA) to effectively find solutions to combinatorial optimisation problems. They provide a framework that allows a GA and DRL to effectively explore the solution space of classical NP-Hard problems, yielding high quality solutions at a low overhead [8]. This work shows the potential of using RL and evolutionary algorithms to effectively explore solution space, which can be beneficial when applied to different types of fitness landscapes.

Considering learning in ALife, RL is one of the less prominent learning algorithms for digital organisms, with supervised learning being the most common [25]. In recent years, different learning algorithms have been combined, highlighting a focus on attempting to model more complex organisms that interact with the environment.

3 Methodology

This section describes the methodology followed, specifically describing the simulation environment, organisms used, brains used and the evolutionary algorithm.

3.1 Simulation Environment

The simulation environments use a modified version of Life Engine². To investigate the effects of within-lifetime learning on evolution, we use a standard world common to both environments. The world consists of different food tiers, each with different food scores which digital organisms gain as the food is eaten. Each food tier has an associated landmark indicating that a particular type of food is nearby. Organisms need to find food to reproduce and they lose energy throughout their lifetime.

The simulation has ticks which represent a timestep in the natural world, with organisms losing one energy value every 10 ticks. There is also a day-night cycle that doubles the rate of energy decay and caves that influence this decay. At night, energy loss is doubled unless in a cave, and during the day energy loss is doubled if in a cave.

For each world in an environment, it is generated using a seed, placing food cells, cave cells and landmark cells deterministically, ensuring consistency when comparing across worlds. Each world has varying density gradations, with each food type moving from high-density to sparse layouts.

For the simulation experiments:

- (1) Baseline environment: organisms need to learn the association between different food tiers and types of food with different food scores.
- (2) Predator environment: organisms need to avoid energy-draining predators that act as a barrier to finding food. These predators are dynamic and move around the world, but behaviour and movement speed are pre-programmed so there is no risk of co-evolution. The absence of co-evolution keeps the fitness landscape stationary as the genomes of the organisms do not need to evolve in response to predator genomes, which would change the shape of the landscape over time. The predators use a Manhattan Distance scan to find possible organisms, attach to the organisms and drain them of their energy per-tick, decreasing the lifespan.

Different environments are used to compare the effects of within-lifetime learning on unchanging and unpredictable environments. In the baseline environment, digital organisms only need to learn the association between landmarks and food types while exploiting the caves. In the predator environment, these organisms need to learn this association while avoiding predators that act as a barrier to gaining food.

3.2 Organism and Network Architecture

Digital organisms have four eye cells that allow sight across the four cardinal directions, mover cells which allow movement and mouth cells which allow them to eat food. The organisms can see 200 cells in all four cardinal directions whereas predators have a radius of 31 cells, allowing organisms to identify predators in advance. Organism eyes encode the nearest non-empty cell. The morphology remains fixed, so only the brains of the organisms can change, being phenotypically plastic. The input size is 54 from the (four directions per eye · 13 cell types + two scalar inputs) with a

hidden layer size of 128 and six possible actions (four movement directions + two rotation directions).

Digital organisms each have a brain represented by a feed-forward (MLP) with one hidden layer with 128 neurons. The state space consists of the different cells in the world such as food type, food tier, cave cells, other organism cells and predator cells. The action space consists of the four cardinal directions and 90-degree rotations in the left and right directions. The initial weights of the brain when the simulation starts are determined using a Xavier initialisation [7] sampled from a Uniform distribution with the weights clipped to $[-1, 1]$.

At the start of a generation, there are 100 organisms that spawn and are able to asexually reproduce if enough food is eaten, and die if they run out of energy. This asexual reproduction only happens within a generation. Organisms need to acquire an energy score of 3.5 in order to reproduce, and there is an 80% chance that reproduction succeeds. If it fails, organisms lose the acquired energy as they would if reproduction was successful. When organisms reproduce within their lifetime, they pass on copies of their starting weights to offspring with a 5% chance of mutation, which is implemented as Gaussian weight perturbation.

3.3 Within-Lifetime Learning

To simulate within-lifetime learning, the MLP uses the Monte-Carlo policy gradient (REINFORCE) algorithm [29] in an online, per-tick Partially Observable Markov Decision Process (POMDP) [3] form with trace decay and running mean reward baseline. Instead of using episodic returns, we use an online trace-based form. This was chosen as REINFORCE chooses the next action from a probability distribution that tightens as agents learn, trace decay was used to simulate short-term memory and subtracting the exponential moving average (EMA) baseline from the reward was used to reduce variance.

Equation 1 shows the per-tick reward r_t , accumulated as

$$r_t = r_{\text{food}} + r_{\text{energy}} + r_{\text{drain}} \quad (1)$$

where

$$\begin{aligned} r_{\text{food}} &\in \{0, 0.01, 0.5, 1, 2\}, \\ r_{\text{energy}} &= -0.05 \cdot \Delta e_t, \\ r_{\text{drain}} &= -0.5 \cdot d_t, \end{aligned}$$

Δe_t is the energy lost to decay at tick t , and d_t is the predator drain applied at tick t .

The reward function scales according to the food score obtained and the energy that organisms lose which change in response to the day-night cycle or predator drains. The 0.05 and 0.5 multipliers ensure that negative rewards are not weighted too heavily in response to the positive rewards from eating food.

Using only energy for the reward function allows organisms to prioritise food-seeking behaviour while not forcing certain behaviours that give an advantage over the non-learning condition.

The learning rate for the baseline environment is 0.01 and 0.02 for the predator environment after doing a log-sweeping across learning rate values $\{0.001, 0.01, 0.1\}$ and more localised sweep across $\{0.005, 0.01, 0.02, 0.03\}$ on each environment. The trace decay γ value is 0.9 as this is a standard default in RL literature [23], which

²<https://github.com/JoshuaPianoGuy/LifeEngine>

equates to ~6.6 ticks. We also performed a sweep across the values {0.9, 0.95, 0.98, 0.99}, finding that 0.9 led to the optimal fitness.

3.4 Evolutionary Algorithm

To simulate natural selection, we use an evolutionary algorithm (EA) that chooses organisms with the highest fitness score as parents for each new organism in the next generation. The EA is only used between generations to determine the brains of the 100 founding organisms of the next generation. Equation 2 shows the fitness function, which is the cumulative food score that an organism has acquired within its lifetime.

$$\text{Fitness}_i = \sum_{t=1}^{T_i} v_{\text{food}}(t) \quad (2)$$

where T_i is the lifetime in ticks of organism i , and $v_{\text{food}}(t)$ is the energy value of any food eaten at tick t .

Unlike energy, the fitness score only increases over the lifetime of an organism, indicating how much food and the quality of food that has been eaten. The values of the food are 0.01 for the default food, 0.5 for the low food, 1 for the medium food and 2 for the prestige food.

For the EA, we use conventional neuro-evolution where changes are applied only to the weights of the MLP and not the topology [19]. We define a genome as the starting brain of each organism, which is used during the crossover step, mirroring non-Lamarckian inheritance as learned weights are never transferred to offspring. The EA uses tournament selection with a k value of 2, followed by a uniform crossover step between parents with a 3% probability of Gaussian weight perturbation which serves as the mutation operator.

4 Experimental Design

This section describes the different experimental conditions to investigate our research questions, the metrics used and the final parameters used in the environment and RL for the final runs.

No ethical considerations had to be made when designing the experiments as there were no living subjects, and the code-base is open source.

4.1 Experimental Conditions

To test the effects of within-lifetime learning on evolution, we have three experiments to isolate the effects of only evolution, isolate the effects of only within-lifetime learning and the effects of both being present. Across these three experimental conditions, the only difference is the presence of either the EA or RL. The three experimental conditions are:

- (A) Only the EA, serving as the evolution condition. Organisms are born with starting weights that do not change within their lifetime. These unchanged weights are used during crossover by the EA.
- (B) The EA and RL, serving as the within lifetime learning condition. Organisms are born with starting weights that change at each tick using the REINFORCE algorithm. During crossover, the starting weights are used, so learning is discarded, mirroring non-Lamarckian inheritance.

- (C) Only RL, where there is no selective mechanism, making this learning without selection and similar to Lamarckian inheritance as changed weights are inherited. This serves as the pure RL condition as the effects of learning are isolated by removing the selection mechanism. To get the 100 founding organisms for each new generation, a buffer of the most recent 100 organisms are kept and if above 100 at the generation boundary, organisms are randomly removed to get to 100 founders. The buffer size of 100 was chosen to make metrics directly comparable to those of conditions A and B. Similarly to the other two conditions, there is a 5% chance of mutation during within-generation reproduction and a 3% chance of mutation between generations.

In each experiment, there are 100 founding organisms at the start of a generation, who eat food, avoid predators, reproduce and die. Each generation is 10 000 ticks, cycling through 5 different world maps for 2 000 ticks per map. At the end of each generation, the evolutionary algorithm selects organisms for crossover which determines the starting weights of the 100 founding organisms for the next generation.

4.2 Metrics

To compare the performance of each experimental condition, we use the mean fitness of each generation normalised to [0, 1] using min-max scaling computed jointly across environment and runs.

To investigate genetic assimilation in the conditions that use RL, we track the mean absolute weight difference over time which is the mean of the difference between weights at the end of organism's life and the beginning of its life. If learning leads to genetic assimilation and the genome expressing learned behaviours becomes genetically encoded, then this value should decrease over time as weights shift less every generation.

Finally, we compare the mean converged fitness in the last 50 generations across conditions to see the average fitness of each run towards the end of the 1 000 generations.

We define the viability threshold in Equation 3 as:

$$\begin{aligned} \text{viability threshold} &= 3.5/0.8 \\ &= 4.375 \text{ fitness} \end{aligned} \quad (3)$$

The viability threshold is defined as the cost of reproduction over the chance of reproductive success, indicating the fitness that a generation needs in order to sustainably reproduce. We normalise this value to [0, 1], but this indicates which runs and conditions produced populations above this viability threshold. Seeing that we normalise the viability threshold for different evaluations that are separate to our main experiments such as those for the landscape slices, time to reach the viability threshold and evaluations of inherited genomes, this normalised value does differ as we use min-max scaling across all measured values.

Related to the viability threshold is measuring the time it takes to reach this threshold for each experimental condition. We use a Kaplan-Meier estimator [12] because censoring is non-informative, which is important as the mean fitness of some runs might still be increasing at generation 1 000. We record the first generation at which the mean fitness is above the viability threshold for 10 generations because from empirical analysis, most runs cross this

threshold, so this would distort the Kaplan-Meier median. Censoring is also administrative, stopping at 1 000 generations as this is constant across all runs.

4.3 Fitness Landscapes

To show differences in the fitness landscapes between the two environments, we adapt and apply the method proposed in Li et al. for visualising a loss landscape [14] to the visualising the fitness landscape. We do this by evaluating the fitness of each possible genome after shifting the weights of the anchor genome. This approach does only provide a slice of the landscape which exists in a higher dimensionality, but differences between the landscapes and metrics can inform that differences exist between the two environments. We provide fitness landscape slices for multiple seeds per environment.

To quantify the differences between the landscape slices, we compute Weinberger’s auto-correlation value which indicates the ruggedness of the landscape [28] and the fitness-distance-correlation (FDC) which indicates the difficulty, or how hard it is to find the highest peaks, of the landscape [11]. As the landscape slices presented are continuous as the genomes are continuous, we applied filter normalisation [14] and used the highest observed fitness value for the FDC calculation as the landscape is not deterministic and fitness values can change.

These changes do present a workable representation of the fitness landscape slices, but are limiting as we compress high-dimensionality in 3-D and use an approximation for the FDC calculation instead of the global maximum which is unknown.

Finally, we also represent the viability threshold on each landscape slice and measure how this changes when genomes are evaluated with and without the presence of RL.

4.4 Experimental Variables

To determine experimental variables, we tested a range of environmental and RL hyper-parameter values using multiple runs across the same seeds as we discovered that the outcome of each run could be different even though the world maps were terrain seeded.

When tuning these parameters, we used the same sized world as in the final experiments but on different terrain seeds to these runs. Across tuning runs we used the same seeds to compare the effects of different parameters on fitness.

4.4.1 Environmental Parameters. Each experimental condition is run on both the baseline and predator environments. The final environmental parameters are found in Table 1.

To determine these, we performed sweeps across different numbers of predators and different per-tick drain amounts. The range for predators was {15, 30, 45, 60} and the range for drain amounts was {1, 3, 5, 10}. We tested these across multiple runs across multiple seeds, and chose values that allowed population and fitness stability, but also acted as a barrier.

4.4.2 Network and Reinforcement Learning Parameters. Different RL hyper-parameters were tested, and the final values are found in Table 2. The hidden layer size of 128 was chosen as the other tested values of 32 and 64 showed runs from the evolution condition plateauing at a lower mean fitness more often. The 128 size

Table 1: Final environmental parameters. The upper block is shared by both environments; the lower block applies only to the predator environment.

Parameter	Value
<i>Shared</i>	
Low food score	0.5
Medium food score	1.0
Prestige food score	2.0
Energy decay per 10 ticks	1
Decay multiplier when exposed	×2
Day and night length	300 ticks
<i>Predator environment only</i>	
Roaming predators	60
Drain per tick on contact	5
Detection radius	31 cells

Table 2: Final network and reinforcement learning parameters, held constant across all conditions that use within-lifetime learning.

Parameter	Value
Hidden layer width	128
Learning rate, baseline environment	0.01
Learning rate, predator environment	0.02
Eligibility trace decay (γ)	0.90
Reward baseline decay, EMA (β)	0.90
Energy decay reward penalty multiplier	-0.05
Predator drain reward penalty multiplier	-0.50

Table 3: Final experimental design used for the reported runs.

Parameter	Value
World size	500 × 500
Generations per run	1000
Map seeds	10
Runs per seed	10
Environments	2
Experimental conditions	evolution, learning, pure RL

was chosen as it favoured the evolution condition by being below the viability threshold the least amount of times. Only learning rate made a difference in performance, with 0.01 being optimal for the baseline environment and 0.02 being optimal for the predator environment.

4.4.3 Final Experiment Parameters. The final experiment parameters can be found in Table 3. For the final experiments, we used seeds that were different from those used during parameter tuning. We opted for more runs and more seeds instead of a larger generation count to show whether the results hold given the per-run stochasticity. We used 10 different seeds to generate the terrain for each world map, 10 runs per seed and 1 000 generations per run, so 10 million ticks.

4.5 Validation & Reliability

To ensure that weights only changed during the conditions that had RL, we made sure that the difference between the weights at the end of the life and beginning of the life, or mean absolute weight difference, was 0. Similarly, we ensured that this difference was non-zero for the conditions that used RL. To ensure that the world maps were identical across conditions, we used a hash function, comparing the hash value ensuring that it was always the same. To ensure that energy is preserved when in caves, we checked that the energy loss was 0, and we also tracked the energy loss per-tick when predators attached to organisms.

To ensure that the results are reliable, we ran each condition 100 times in total on each environment. This is 300 runs in total per environment, so 600 runs in total to account for the per-run stochasticity. These runs were done on different world map seeds than the ones used during parameter tuning to ensure that initial observations still held on different seeded terrains.

5 Results & Discussion

This section presents our results that address our research questions. For this section, the evolution condition is referred to as condition A, the learning condition is referred to as B and the pure RL condition is referred to as C.

5.1 Fitness Results

Figure 1 shows the normalised mean fitness values over 1 000 generations for all three conditions in each environment. In the baseline environment on the left, both conditions A and B reach roughly the same mean fitness score of ~ 0.84 with condition C being lower. All three conditions plateau and conditions A and B have nearly identical mean fitness over 1 000 generations. Condition C also plateaus, but at a lower fitness score. The shadows of each graph show the spread of data over all runs, and in the baseline environment, condition C has the highest standard deviation of 0.014 as opposed to ~ 0.006 and ~ 0.008 of conditions A and B respectively.

When the predator environment is considered on the right, condition B has the highest mean fitness score of 0.8 by generation 1 000, followed by condition A with a score of 0.74. The mean fitness of condition A does increase and approaches the score of condition B, but is below conditions B and C for 700 generations. Condition A also has the highest spread of data with a standard deviation of 0.124. Condition B has the lowest spread of data with a standard deviation of ~ 0.007 , showing that performance is consistent across all runs in the predator environment. Condition C has a higher standard deviation of ~ 0.059 , and has a similar shape to that of condition B, plateauing at a score of 0.68.

Across all conditions, the mean fitness dropped between the two environments, but conditions A and B had lower decreases than condition C. By generation 1 000, the mean fitness across all runs is above the viability threshold. It is important to note that the mean fitness conceals any runs that under-perform as it averages across runs.

For our discussion of Figure 1, we only consider the mean fitness values. We consider the spread of data further in Section 5.2 within-lifetime learning has no positive impact in the baseline environment where the task is straightforward. Both conditions A and B reach

the same mean fitness peaks with little variance, showing that evolutionary search suffices to climb the fitness landscape as quickly as when within-lifetime learning is present. Although the fitness of condition C is lower than that of A and B, the performance is more consistent than in the predator environment. This suggests that in an easy environment, the absence of a selective mechanism has no negative impact on climbing the fitness landscape and suggests that the presence of within-lifetime learning suffices when traversing the landscape, although this traversal is less effective than evolutionary search. These results in the baseline environment disprove the hypothesis that learning is beneficial in both environments.

When considering the predator environment, condition B has the best mean fitness values which shows that within-lifetime becomes beneficial when there is an active barrier to increasing fitness. The mean fitness of condition B is consistently above that of conditions A and C, showing that in the presence of a selective mechanism, within-lifetime learning achieves the highest fitness scores. This is supported by the mean fitness of condition C being above that of condition A for 700 generations: the presence of within-lifetime learning increases fitness even in the absence of a selection mechanism. Although evolution suffices in later generations, within-lifetime learning seems to reach higher fitness scores more quickly on average.

Figure 1 shows that within-lifetime learning consistently reaches fitness scores more quickly on average, supporting RQ2.

5.2 Fitness Variance

Figure 2 shows the spread of the fitness values across all runs in both environments. Each point is the mean fitness of the final 50 generations, showing the final fitness values that runs converged to. This figure helps explain the high standard deviation of conditions A and C in the predator environment, showing the large spread of final fitness scores across runs.

In the baseline environment, conditions A and B are tightly clustered with condition C being more spread out. Across all conditions, the mean and median fitness of each condition are roughly identical, being overlaid on one another. The predator environment, however, shows that the clustering in condition B is nearly identical to that of condition B in the baseline environment, but conditions A and C are more spread out. Condition B has a unimodal distribution with the mean and median being nearly identical. Condition A is bimodal, with most runs clustering around the median of 0.79 but the mean being 0.74. There are also 10 runs in condition A that do not reach the viability threshold by the end of generation 1 000. Condition C also has a bimodal distribution with the mean being 0.68 and the median being 0.7. There are 2 runs in condition C that do not reach the viability threshold by the end of generation 1 000. In these conditions, most runs cluster at higher fitness scores, but the mean is brought down by the runs that underperform.

The strip plots in Figure 7 in the appendix and the Intraclass Correlation Coefficient (ICC) and standard deviation values in Table 4 show the spread of data for each seed and the effects of the terrain on the outcome. σ_{between} is the standard deviation attributable to map identity and σ_{within} that of replicate runs on a fixed map, both estimated by one-way random-effects ANOVA. The ICC is the variance ratio between the two.

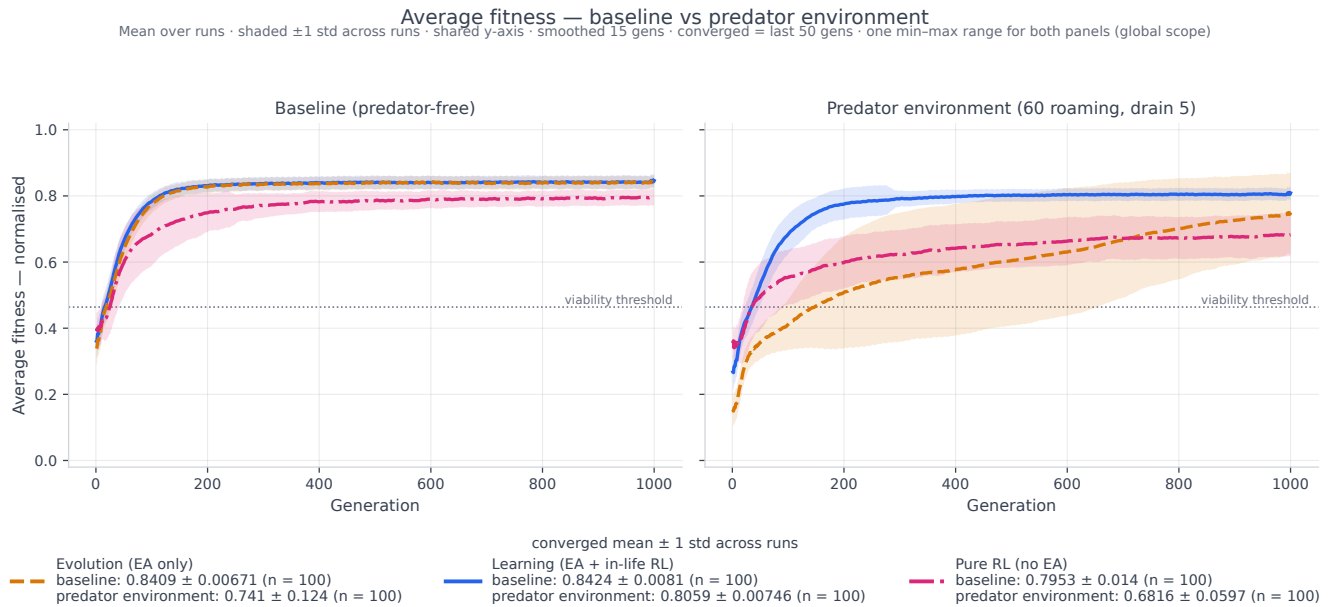


Figure 1: Normalised graph showing the mean fitness of all runs per generation for 1000 generations

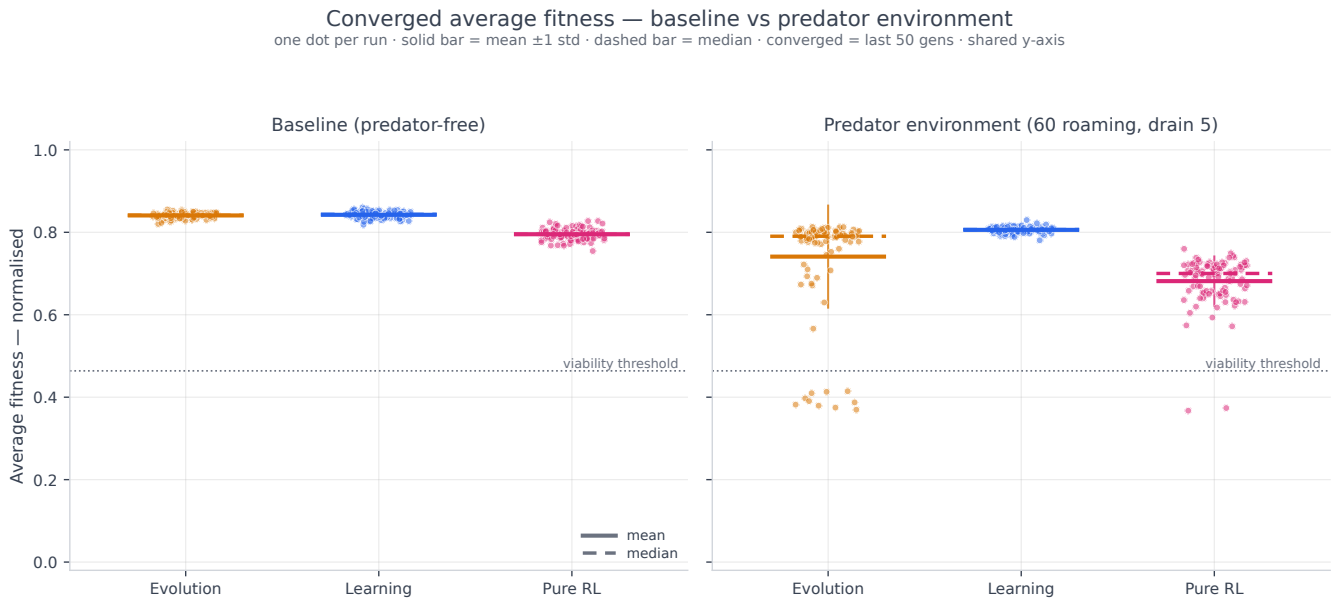


Figure 2: Normalised strip plot showing the mean fitness over the last 50 generations for each run across all conditions in both environments

Considering both the strip plot in the figure and table, there is low variance in the baseline for conditions A and B, while condition C has a higher variance which is supported by the lower ICC value and the higher within-seed standard deviation, showing that the terrain has less of an effect on the variance and that variance happens between runs. In the predator environment, conditions A

and C both have low ICC values with higher within-seed standard deviation, showing that the unpredictable variables like predator movements, mutation and food regeneration contribute to high variance. Interestingly, condition A also has a high between-seed standard deviation showing that there is variance between different seeds, which is unique to this condition.

Table 4: Between- and within-seed variability by environment and condition.

Environment	Condition	σ_{Total}	σ_{between}	σ_{within}	ICC
Baseline	Evolution	0.0067	0.0048	0.0049	0.49
Baseline	Learning	0.0081	0.0044	0.0069	0.29
Baseline	Pure RL	0.0140	0.0025	0.0138	0.03
Predator	Evolution	0.1237	0.0342	0.1194	0.08
Predator	Learning	0.0075	0.0046	0.0060	0.37
Predator	Pure RL	0.0597	0.0063	0.0594	0.01

Across these environments, the ICC and standard deviation of condition B do not change much, showing that the variance of these runs are low across seeds and across different runs per seed, despite the introduction of predators which seems to introduce variance into runs from conditions A and C.

Considering the viability threshold in Figure 2, all three conditions are above in the baseline environment, showing that in an easy environment, evolutionary search leads to a population that can sustain itself. In the predator environment, however, conditions A and C do not always pass this threshold. The distribution of data in conditions A and C is bimodal with runs either crossing the viability threshold or not. Condition C has more runs that do not cross this threshold, showing that evolutionary search is less consistent than only learning with no selection mechanism in an unpredictable environment. This happens as there having no phenotypic plasticity cannot allow organisms to adapt behaviours in dynamic environments. Condition B has the tightest clustering and consistently reaches fitness scores above the viability threshold. This shows that even in the presence of predators that introduce variability, within-lifetime learning and a fitness-based selective mechanism can reliably increase fitness. From the predator environment, it seems that the presence of within-lifetime learning is beneficial when there is difficulty in the environment. Even in the absence of a selective mechanism, learning crosses the viability threshold more reliably, even at the cost of lower fitness. These results support RQ1 as the presence of within-lifetime learning helps populations reach viability, especially in dynamic and more difficult environments. Additionally, they show that within-lifetime learning is only has a measurable benefit when phenotypic plasticity is necessary to survive.

5.3 The Effects of Learning on Genomes

The graphs in Figure 3 show the mean fitness evaluation of the genomes of the 100 founder organisms every 250 generations in both environments. These are the inherited genomes evaluated with no within-lifetime learning. In the baseline environment, the genomes from condition B have a slightly lower mean fitness until generation 500. After this, the genomes from both conditions have similar evaluations and reach roughly identical fitness values at generation 1 000. In the predator environment, however, the mean fitness of genomes from condition A are below that of condition B until generation 1 000. As this is the mean, the extreme values are masked, but at generation 250, the highest fitness from condition A is below the lowest fitness from condition B. Comparing the graphs,

the predator environment has a higher variance, but the genomes from condition B has a lower variance than those from condition A.

Additionally, Figure 4 shows the number of generations it takes for each condition to reach the viability threshold using a Kaplan-Meier estimator, with 1 showing that all generations have passed this threshold. Using the Kaplan-Meier median, the graph shows that in the baseline environment, conditions B and C reach the viability threshold more quickly than condition A, but this difference is small. In the predator environment, however, condition B reaches the viability threshold shortly after condition C, but ~7.9 times sooner than condition A.

The evaluation of inherited genomes graph in Figure 3 reveals that evolution outperforms the learning condition on the baseline environment in terms of mean fitness, being higher until generation 500. This suggests that learning is detrimental to performance and carries a cost, which was also suggested by Mayley who said that learning carries a higher cost than inherited behaviours [16]. Despite this, the mean fitness scores of both conditions in the baseline environment are roughly the same, showing that although the genomes from the learning condition are worse until generation 500, within-lifetime learning masks this, also supporting another suggestion by Mayley [17]. The evaluations from the predator environment, however, show that after 250 generations, the genomes from the learning condition have a higher mean fitness than those from the evolution condition. Despite the higher variance, the lowest fitness score from the learning condition is still greater than the highest fitness score from the evolution condition. This suggests that fitness-increasing behaviour becomes innate more quickly when within-lifetime learning is present. Up to generation 480, all genomes from the learning condition have a higher fitness than those from the evolution condition. These results suggest that within-lifetime learning can speed-up genetic assimilation, which supports the Baldwin Effect. These results support our prediction that evolution encodes behaviours less quickly than learning, and also partially disprove our prediction that learning needs a selective mechanism to be useful. Even with no selective mechanism, behaviours become encoded quickly which allows for speed at the cost of fitness.

The time to reach viability graph in Figure 4 reveals interesting results, showing that in the baseline environment, all three conditions reach viability early on, whether or not there is a selective mechanism. In the predator environment however, the benefits of within-lifetime learning are evident as the learning and pure RL conditions reach viability hundreds of generations before the evolution condition. This suggests that the presence of learning helps to reach the viability threshold more quickly, which hints at behaviours becoming genetically assimilated more quickly.

These results support both research questions, showing that within-lifetime learning reliably reaches viability more quickly while also accelerating genetic assimilation in the predator environment and also supports our hypothesis that evolution encodes behaviours less quickly than within-lifetime learning.

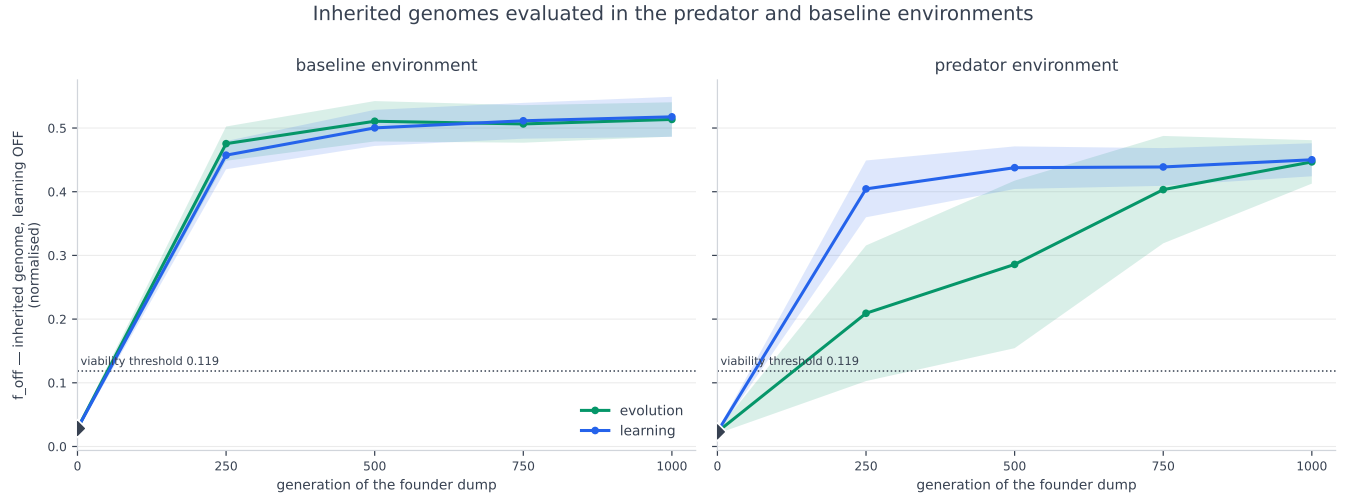


Figure 3: Graphs showing the mean fitness evaluations of the inherited genomes every 250 generations across the baseline and predator environments

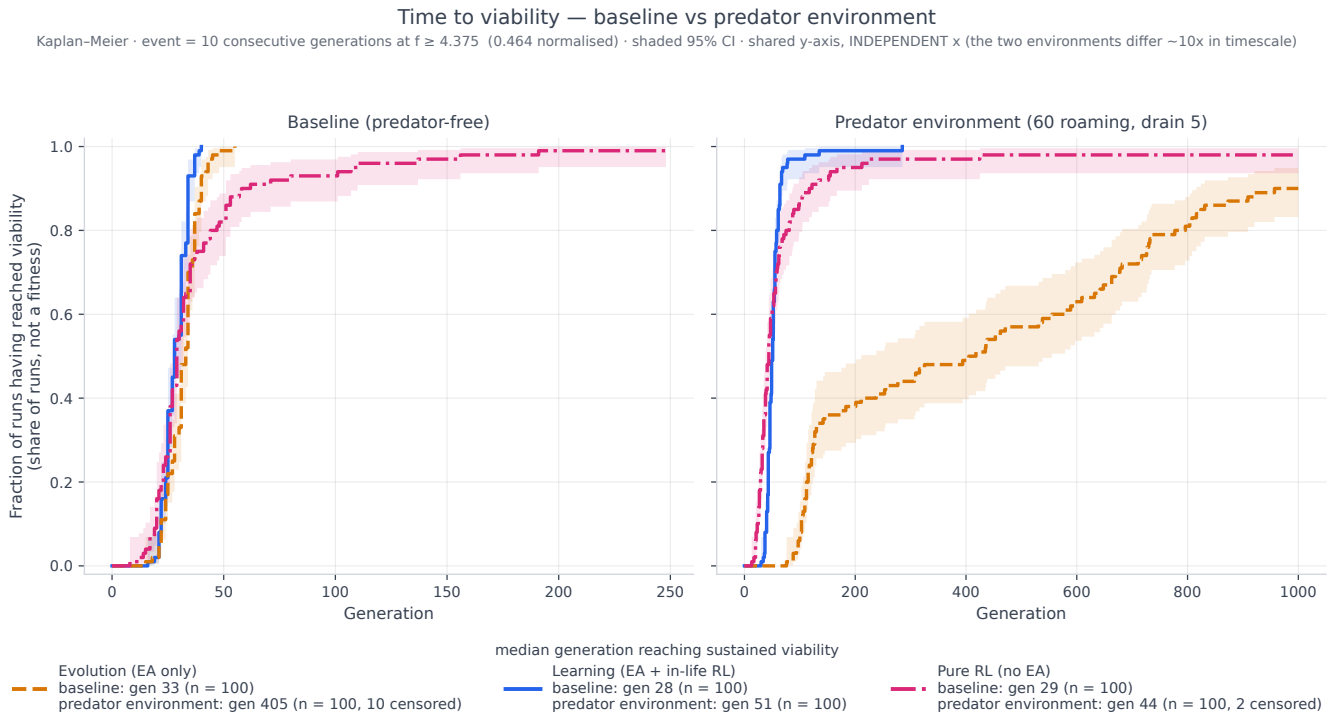


Figure 4: Graph showing after how many generations each condition crosses the viability threshold in both environments

5.4 Fitness Landscape Slice Results

The fitness landscape slices can be seen in Figures 5 and 6, showing views of fitness landscapes across seeds with Principal Component Analysis (PCA) trajectories [10] from runs overlaid on the landscapes. On the left-hand side is the landscape construction from genomes with no within-lifetime learning, and the right-hand side

shows landscape constructions where within-lifetime learning is enabled.

In the baseline environment, the landscape slice is viable nearly everywhere, and when within-lifetime learning is enabled, this viability increases to nearly 100% if below. All runs from the baseline environment also end up in regions of high fitness as shown by

the PCA trajectories, showing that the genomes are able to reach regions of high fitness. In the slices from the predator environment, however, runs where genomes from condition A do not reach the viability threshold, the landscape is only partially viable. When the same landscape is constructed and within-lifetime learning is enabled, the landscape becomes more viable. The PCA trajectories show that genomes from condition A land in regions of low fitness when below the viability threshold, but also how these genomes land in regions of higher fitness when they perform as well as genomes from condition B.

Table 5 shows the FDC, auto-correlation length (λ) and fraction of the landscape that is above the viability threshold computed using the slices from multiple seeds per landscape. λ is roughly the same across both environments, with the FDC of the fitness landscape of the baseline environment sitting at -0.7 and the FDC of the landscape from the predator environment having portions where the FDC is more negative and less negative. The viability fraction of the landscape from the baseline environment is high, nearly reaching 1, and the fraction of the landscape from the predator environment is lower. Across these metrics, the metrics from the predator environment have a higher standard deviation than the metrics from the baseline environment.

The fitness landscape slices in Figures 5 & 6 show that when the landscape is constructed when within-lifetime learning is enabled, the landscape is lifted into regions of higher viability. In the baseline environment, this effect is marginal, but in runs where evolution does not reach the viability threshold in the predator environment, the effect is more pronounced with viability nearly doubling in one seed. This supports the suggestion made by Hinton & Nowlan that learning smooths the fitness landscape. This observation came from an extreme fitness landscape with only one large peak, so this lifting is more subtle, but genomes on the landscape move into regions of higher fitness when within-lifetime learning is present. The metrics from the landscape in Table 5 show that the fitness landscapes of both environments are smooth as the auto-correlation, shown by λ , is roughly the same. The FDC values for both environments are close to -1, suggesting strong correlation as genomes are close to the best observed genome, but the region of the landscape when evolution runs fail is slightly more positive moving towards the "difficult" problem class. As by Jones & Forrest, if the FDC is near -1 then fitness rises as distance to the global maximum decreases, creating an easily searchable landscape [11]. The lowest FDC value of -0.58 is from the predator landscape, suggesting that there are regions on the predator landscape that are more difficult than in the baseline environment. The viability fraction suggests that in the predator environment, there are regions of low fitness that do not exceed the viability threshold but nearly all of the landscape is viable in the baseline environment. Although we cannot confidently say that the predator environment is multi-peaked, it has higher variance in its metrics, particularly the viability fraction which suggests that the landscape is not uni-modal as there are regions of high and low fitness. As λ is nearly identical suggesting a smooth landscape, learning is beneficial even on smooth yet more difficult landscapes, despite suggestions that landscapes must be rugged for within-lifetime learning to be beneficial from Borenstein and Mayley [4, 17]. Considering the fitness values recorded and the

Table 5: Fitness–distance correlation, correlation length and viable fraction for each environment, computed from multiple landscape slices (mean $\pm$ SD across slices).

Slice	FDC	λ	Viable fraction
<i>Baseline</i>			
Strongest	-0.727 ± 0.101	10.80 ± 0.28	0.997 ± 0.004
Weakest	-0.761 ± 0.099	10.15 ± 0.88	0.994 ± 0.009
<i>Predator</i>			
Strongest	-0.891 ± 0.078	10.04 ± 0.47	0.816 ± 0.175
Weakest	-0.588 ± 0.142	10.40 ± 0.17	0.466 ± 0.272

landscape metrics, learning seems to reshape the fitness landscape into one with regions of higher fitness.

6 Conclusions

From these experiments, we show that within-lifetime learning is a robust mechanism to consistently increase fitness when the fitness landscape is more difficult and has regions of high and low fitness. If the environment is smooth and not difficult, then natural selection suffices when increasing fitness. Even on such a landscape, there needs to be a fitness-based selection mechanism to effectively climb the landscape as continuous learning cannot climb the landscape as effectively.

Interestingly, within-lifetime learning becomes beneficial when the environment changes into one where there is a barrier to increasing fitness. The benefit of learning is seen on smooth yet more difficult fitness landscapes, allowing genomes to climb to high regions of fitness. Within-lifetime learning also increases the rate at which fitness-increasing behaviours become encoded in earlier generations, even if natural selection is able to do the same given more time. Even in the absence of a selective mechanism, learning allows organisms to traverse the fitness landscape differently than natural selection only where genomes remain fixed. We also suggest that within-lifetime learning increases the rate of genetic assimilation, showing the Baldwin Effect.

Within-lifetime learning also reshapes the fitness landscape by providing a lift into higher regions of fitness. When there is plasticity, the genomes can more quickly inherit beneficial learned behaviours, lifting the fitness landscape.

Despite these results, there are limitations with regards to the simulation and environments we used. Our environments, although biologically inspired, are not as complex as those found in nature where there can be multiple species, ecosystems and predators. Additionally, our experiments are not fully emergent as we designed the environments and use an EA, which is an explicit rule. Future work could investigate a more diverse range of environments as well as try to design experiments to be more emergent by enforcing fewer programming rules.

Acknowledgments

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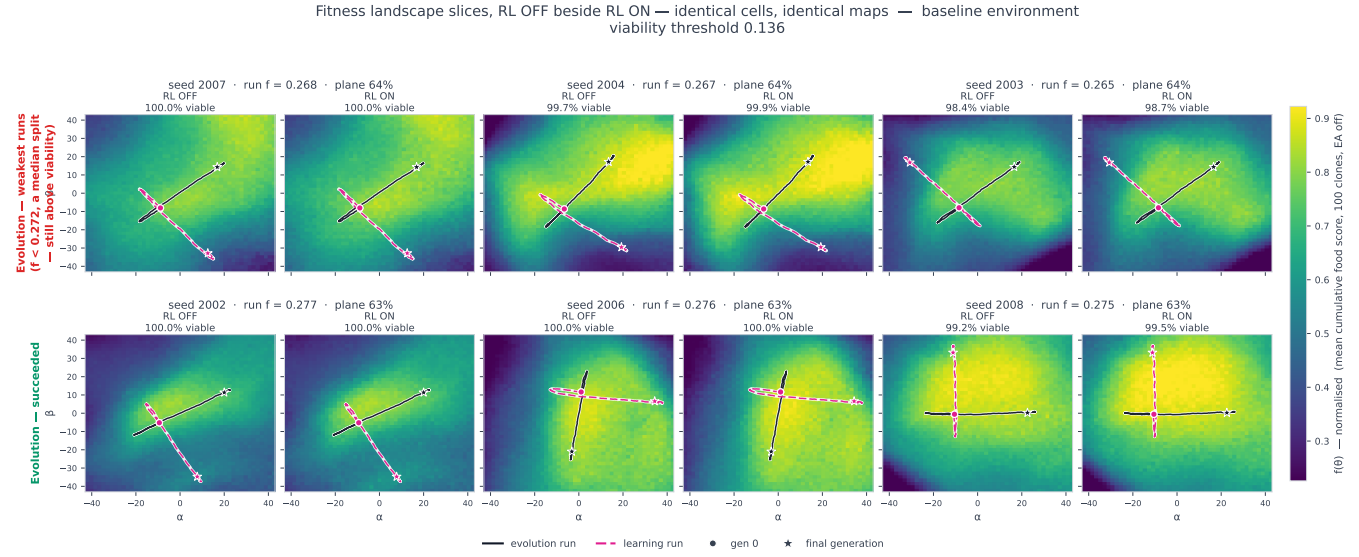


Figure 5: Fitness landscape slices of different world maps from the baseline environment with a PCA trajectory of the evolution and learning runs overlaid

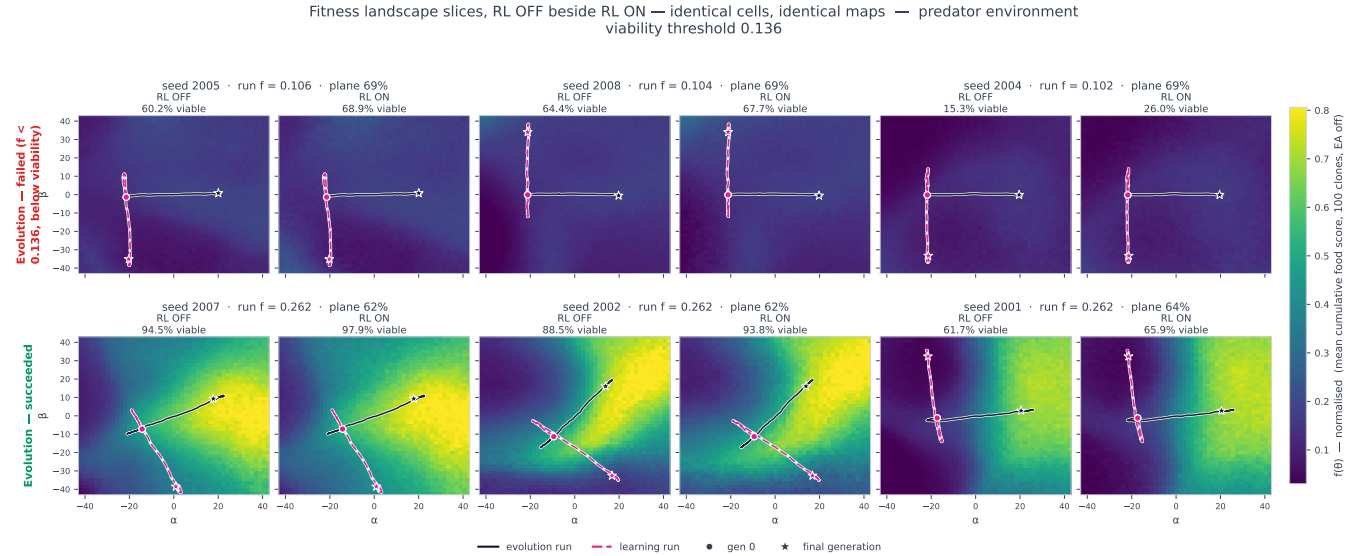


Figure 6: Fitness landscape slices of different world maps from the predator environment with a PCA trajectory of the evolution and learning runs overlaid

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AI Usage Declaration

Anthropic’s Claude Opus 5 was used to check grammar, content and validity of claims in relation to existing work. Suggestions were considered and excluded where not useful or relevant to the paper. Claude was also used to restyle the tables and fix an issue with table spacing.

A Supplementary Results

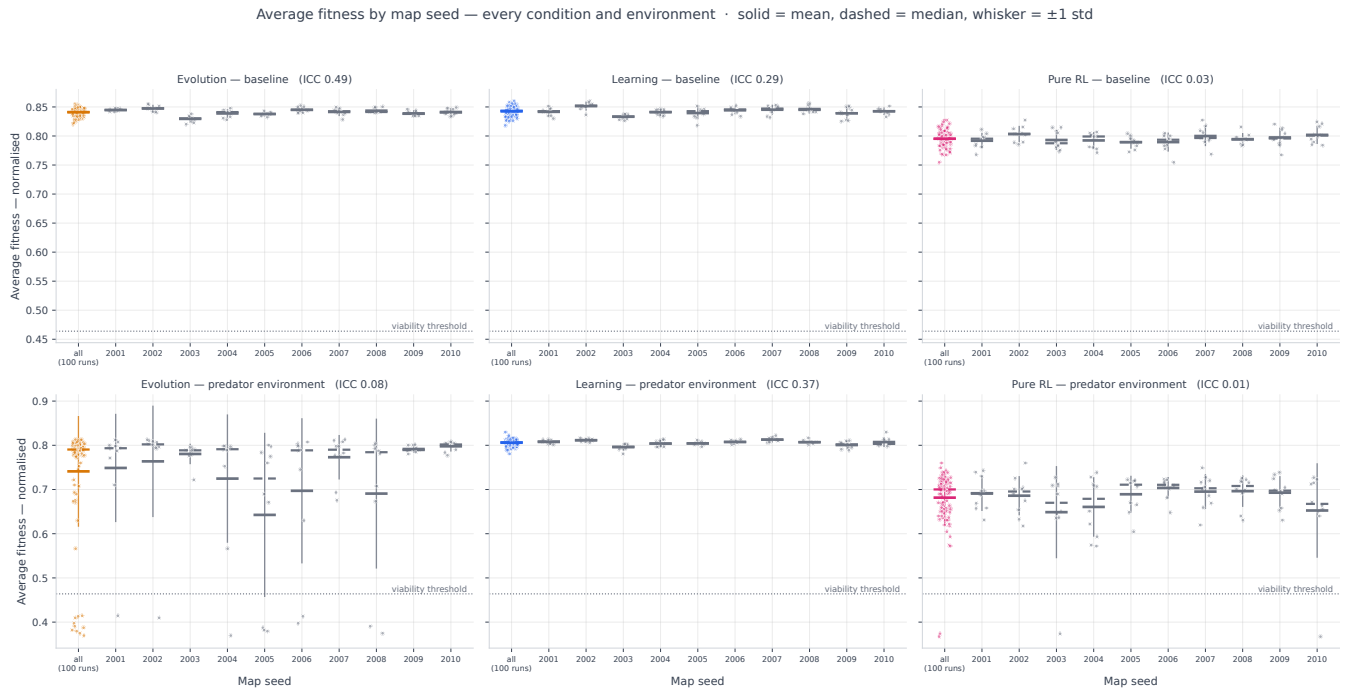


Figure 7: Normalised strip plots showing the mean fitness over the last 50 generations for each seed across all runs in each environment